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3041 | Course Overview | Mini Formula Bank | Module 1 | Module 2 | Module 3 | Module

Diploma in Electronics / Electronics & Communication Engineering • Semester 3

Electric Circuits & Networks

Full-width desktop lesson sample with textbook-style explanation, diagrams, formula cards, Malayalam support, module-wise questions, and a matching Master Answer Key. Only one selected section loads on screen.

Course Code: 3041 | Category: Program Core | Credits: 3 | L:T:P = 2:1:0 | Theory Subject

Lesson structure check

4
Official modules

45
Periods / semester

A4
Print-ready layout

1-click
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Course Overview

Subject title: Electric Circuits & Networks

Semester: 3 | **Course category:** Program Core | **Subject type:** Engineering Theory subject

This subject connects basic electrical laws, AC circuit behavior, network analysis, transformers, DC machines and AC machines. A student who studies this subject properly can read simple circuit diagrams, calculate circuit quantities, apply network theorems, and explain the working of common electrical machines used in electronics and industry.

Malayalam support: ഈ subject-ൽ electric circuit-ന്റെ behavior, AC signal, RLC circuit, network theorem, transformer, DC machine, AC machine എന്നിവ പഠിക്കുന്നു. Technical terms English-ൽ തന്നെ ഓർത്താൽ exam answer എഴുതാൻ എളുപ്പമാകും.

Source basis: Official SITTTR Revision 2021 syllabus content for course code 3041.

Course objective

To develop the skill to diagnose and rectify electric circuit networks and understand the operations of electrical machines.

Simple meaning: Circuit fault എങ്ങനെ കണ്ടെത്താം, circuit values എങ്ങനെ calculate ചെയ്യാം, motor/transformer working എങ്ങനെ explain ചെയ്യാം — ഇതാണ് main aim.

Prerequisite knowledge

- Basic Engineering Mathematics principles from Mathematics I & II.
- Magnetism concepts from Applied Physics I & II.
- Ohm’s law, units, algebra, trigonometry and simple vector/phasor idea.

Course outcomes

CO	Outcome	Duration	Level
CO1	Solve a given AC circuit to find various parameters using AC signals and component behavior.	9 hours	Applying
CO2	Apply network theorems for simplifying circuits and illustrate transformer operation.	14 hours	Applying
CO3	Illustrate principles and operations of DC machines.	10 hours	Understanding
CO4	Illustrate principles and operations of AC machines.	10 hours	Understanding

Module map

Module 1

AC signals, R-L-C behavior, phasor diagram, AC power, resonance, series and parallel RLC circuits.

Module 2

Ohm’s law, Kirchhoff’s laws, mesh and node analysis, network theorems, transformer principle and losses.

Module 3

DC generator, DC motor, back emf, starter, 3-point starter, DC motor comparison and characteristics.

Module 4

Alternator, AC motor classification, stepper motor, servo motor, universal motor, single-phase and three-phase induction motor.

How to study this subject

1. First learn definitions and units: RMS value, average value, frequency, reactance, impedance, power factor, resonance.
2. Draw every circuit and machine diagram in a separate notebook. Label current direction, voltage, winding, core and load.
3. For numerical problems, always write Given → Formula → Substitution → Calculation → Final answer with unit.
4. For theorem questions, write statement, circuit before simplification, circuit after simplification, calculation steps and result.
5. For machine questions, write principle, construction, working, equation, characteristics and applications.

Exam preparation method

Short answers

Definitions, symbols, units, laws, types, applications and advantages.

Problem solving

Practice AC power, impedance, resonance, Thevenin equivalent, maximum power transfer and transformer ratio.

Long answers

Use diagram + working + equation + application format. This gives better marks.

Malayalam tip: Diagram ഉള്ള question ആണെങ്കിൽ diagram ആദ്യം വരയ്ക്കുക. പിന്നെ working principle എഴുതുക. Formula അവസാനം അല്ല, explanation-ന്റെ ഇടയിൽ കൊടുക്കുക.

Quick navigation

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Desktop layout uses 98vw up to 1760px, so it fills laptop/desktop screen instead of staying small.

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